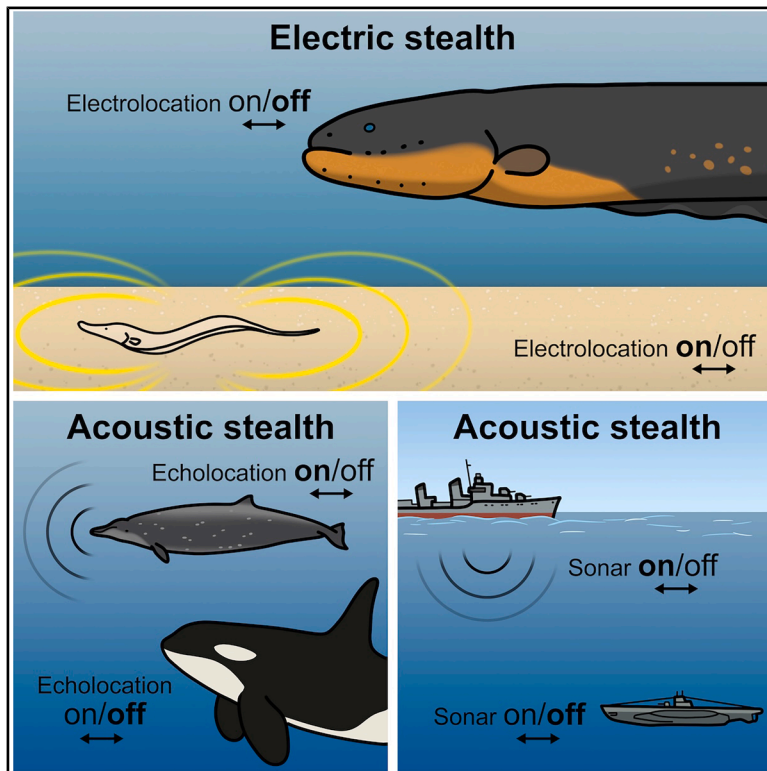


Current Biology

Electric stealth through bidirectional signal suppression in electric eels and their knifefish prey

Graphical abstract



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In brief

Poon and Crampton report bidirectional suppression of electrolocation during encounters between wild electric eels and their weakly electric knifefish prey. Predator and prey alternate between signal emission and silence to balance active sensing with concealment. This electric stealth dynamic parallels acoustic stealth in killer whales and their echolocating toothed-whale prey.

Highlights

- Electric eel signals trigger evasive movements in knifefish in the wild
- Knifefish suppress electrolocation in response to natural and playback eel signals
- Low-frequency energy in eel signals drives knifefish signal suppression
- Electric eels pause electrolocation before high-voltage attacks, then resume



Article

Electric stealth through bidirectional signal suppression in electric eels and their knifefish prey

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SUMMARY

Animals that use signal-based active location systems—electrolocation and echolocation—emit self-generated signals during foraging to sense their surroundings, but these signals are conspicuous to predators and prey. How such signals are modulated to balance sensory sampling against detection risk remains poorly understood. Both electric eels (*Electrophorus*) and their weakly electric knifefish prey are electroreceptive and generate electric organ discharges (EODs) that are essential for nocturnal foraging yet mutually detectable. Using autonomous electric-signal loggers in an Amazonian stream, we recorded signaling interactions between wild electric eels and knifefish. We found that both predator and prey temporarily reduce detectability by pausing electric signaling. Knifefish respond to eel EODs by evasive movements or by switching off their own EODs until the threat subsides. Laboratory-based playback experiments showed that EOD switch-off behavior is phylogenetically widespread among knifefish with pulse-type discharges and driven primarily by the low-frequency energy of eel EODs. Electric eels, in turn, frequently pause low-voltage electrolocation EODs during hunting, allowing them to approach electroreceptive prey with reduced detectability. During these silent intervals, eels forgo active electrolocation, likely relying instead on mechanoreception and passive electroreception. Electrolocation resumes immediately after emitting a high-voltage predatory burst—used for attack or probing—when prey localization outweighs the benefits of concealment. This bidirectional suppression of electric signaling parallels acoustic stealth dynamics in killer whales and their echolocating odontocete prey and mirrors alternating active-passive strategies used by human sonar and radar operators—revealing convergent solutions to the challenge of signals that are essential yet intrinsically conspicuous.

INTRODUCTION

Predator-prey encounters often impose a trade-off between using conspicuous signals and avoiding detection by predators or prey.^{1,2} We show that this trade-off has given rise to a form of bidirectional signal suppression between electric eels (*Electrophorus*) and their weakly electric knifefish prey, both members of the nocturnal Neotropical order Gymnotiformes.³ These fish rely on active electrolocation for orientation and foraging in darkness, sensing nearby objects through distortions of electric fields generated by their electric organs.^{4,5} However, given their shared electrosensory capabilities, eels and knifefish can detect each other's electric signals during close-range encounters. This mutual detectability creates a fundamental conflict: the signals required for active electrolocation can also reveal the sender's location to predators or prey.

One way that animals can reduce detectability is by suppressing signal production (i.e., silencing signals). Such suppression is widespread in communication systems, where silencing calls, displays, or vibrations can reduce the risk of attracting predators,

prey, or competitors.¹ In these contexts, signal suppression interrupts communication without eliminating sensory input from the environment. By contrast, in active sensing systems, animals emit self-generated acoustic, electric, or (rarely) bioluminescent signals^{6–8} during foraging to locate nearby objects, prey, or other animals, thereby coupling signal production directly to spatial localization. Signal suppression in active sensing systems therefore reduces conspicuousness but also limits an animal's ability to monitor its surroundings, including the detection of predators or prey, making silence inherently more costly.⁸

In this paper, “active sensing” refers specifically to such signal-emission-based active location systems and excludes phenomena sometimes described under broader uses of the term that do not involve self-generated location signals. These include movement-based modulation of sensory input and exploration-exploitation strategies implemented without signal emission.^{9,10}

Because suppressing signal emission in active location systems directly compromises spatial perception, animals that rely on active sensing typically generate signals at high duty cycles (i.e., signals are emitted for a large fraction of the time)



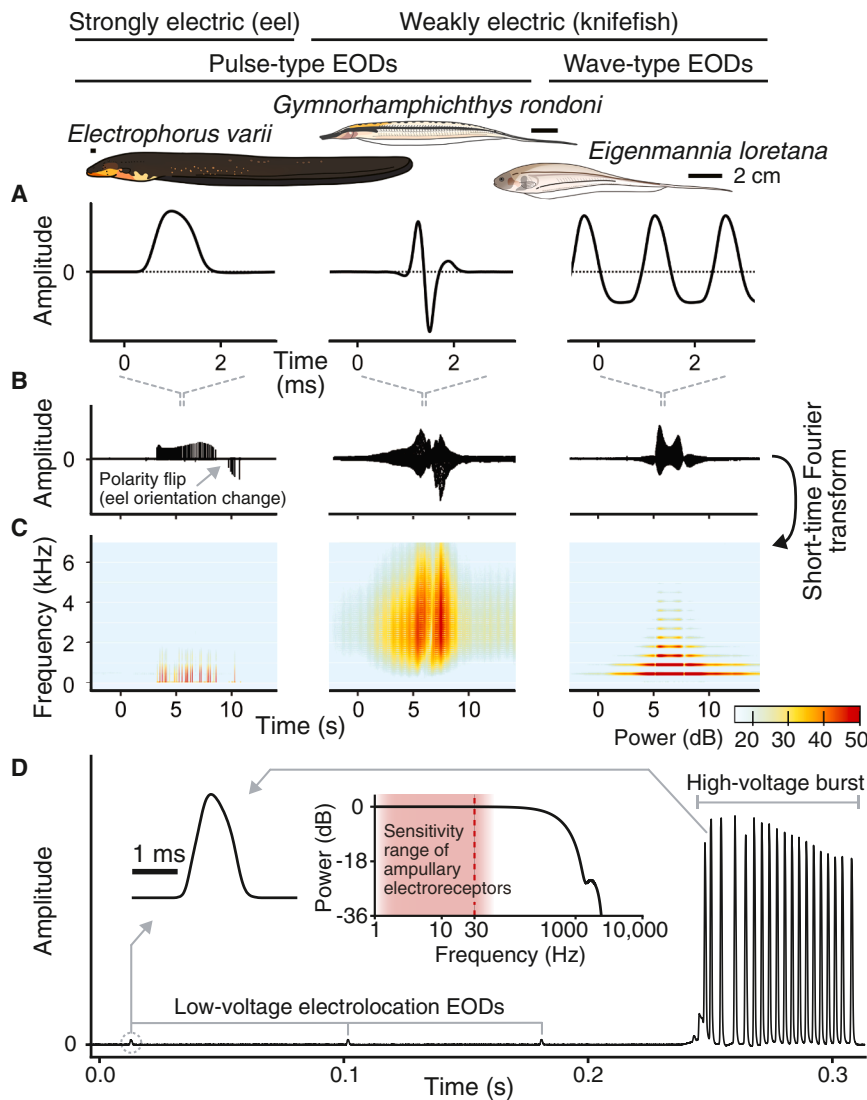


Figure 1. Electric signals of the electric eel and two knifefish species

(A) Electric organ discharge (EOD) waveforms. (B) Example logger recording of a fish passing through the detection range of an electrode pair (a “transit”; STAR Methods). (C) Spectrograms from (B). (D) Eel hunting sequence showing low-voltage electrolocation EODs (EEs) followed by an attack volley (prolonged high-voltage burst). Insets show a representative EOD waveform (left) and power spectral density (right), shared by low- and high-voltage discharges. The dashed line at 30 Hz indicates the peak sensitivity of gymnotiform ampullary electroreceptors.

same signal-emission-based active sensing modality and are mutually detectable through those signals. In these systems, suppressing signal emission reduces detectability but also compromises spatial localization and close-range tracking of objects or prey. This shared constraint leads to a specific prediction: selection should favor context-dependent, transient suppression during periods of elevated detection risk—via on-off regulation—rather than sustained silencing or uninterrupted signaling. The expectation, therefore, is intermittent suppression during predator-prey encounters.

To test this prediction in the electro-sensory domain, we combined two approaches: long-term autonomous logger recordings of electric organ discharges (EODs; Figure 1) in an Amazonian stream and controlled laboratory experiments. Field recordings allowed us to quantify

during a foraging bout, particularly at critical moments such as prey localization. For instance, some odontocetes, such as harbor porpoises, echolocate almost continuously while foraging,¹¹ and foraging bats likewise depend on near-continuous echolocation to track prey and avoid obstacles.¹² Although some bat species briefly silence calls to reduce jamming or avoid detection by competitors,^{13,14} complete signal suppression during foraging is uncommon in these systems.

While active sensing generally demands near-continuous signal production during foraging, a notable exception occurs in transient (Bigg’s) killer whales and their odontocete prey. When stalking ultrasound-sensitive mammals, these killer whales temporarily suppress both echolocation and communication calls,^{15–17} reducing detectability to prey that can hear these signals. In turn, upon detecting killer whale signals, odontocete prey—including sperm whales and beaked whales—take evasive action or minimize echolocation and social calls, resuming them once the threat subsides.^{18–21}

The eel-knifefish system shares a key feature with the killer whale-odontocete system: both predator and prey rely on the

natural encounters between hunting electric eels and knifefish, including prey movements and predator or prey EOD cessations. Laboratory playback experiments then isolated the sensory cues driving prey EOD cessation and assessed their generality across knifefish families. Together, these approaches supported the prediction that predator-prey interactions in this system involve context-dependent on-off modulation of electric signaling: knifefish reduced or ceased EOD production when eels were nearby, and eels suppressed electrolocation before emitting high-voltage predatory bursts.

RESULTS

EOD logger system and field recordings

We deployed six custom-designed EOD loggers along a 150 m transect in a small clear-water Amazon rainforest stream (width ~4–5 m, mean midstream depth 0.4 m) (Figure S1). Each logger used two orthogonal electrode pairs for omnidirectional electric field detection, with the electrodes anchored to the substrate (Figure 2B). Signals were amplified, digitized,

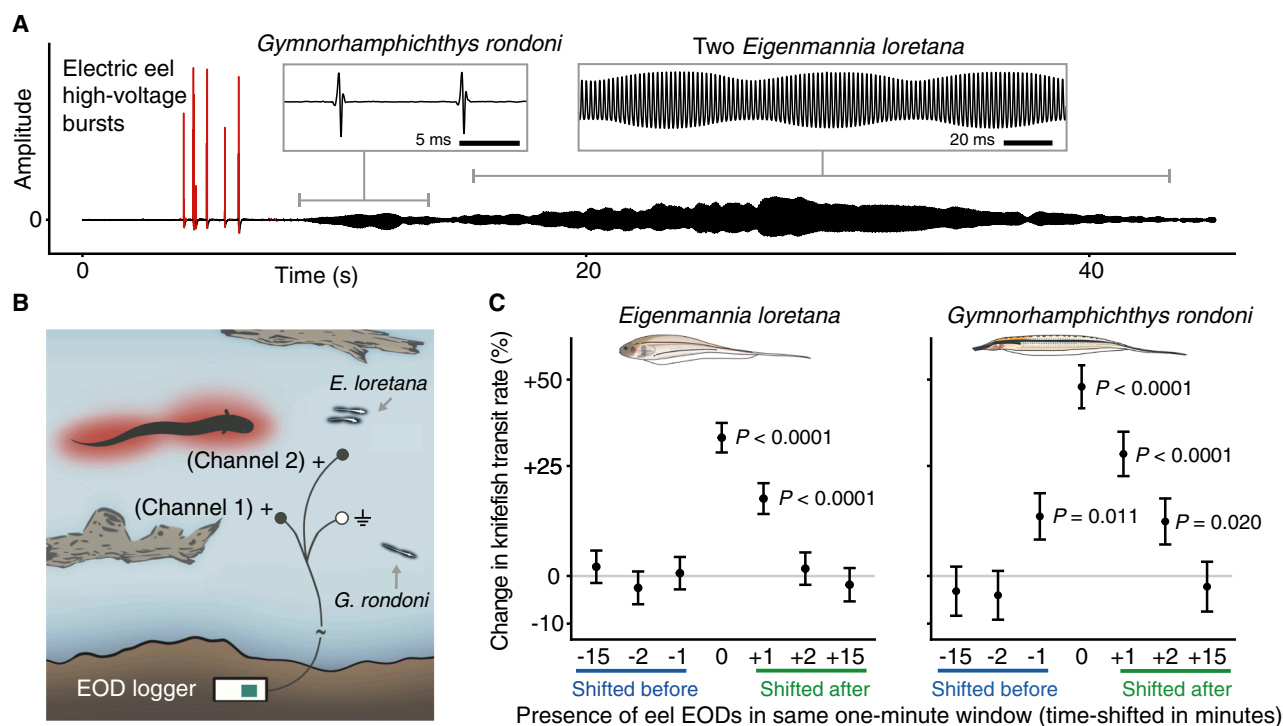


Figure 2. Knifefish exhibit evasive movements in response to electric eel EODs

(A) Logger recording showing high-voltage eel bursts (red) followed by transits by *Gymnorhamphichthys rondoni* (pulse-type EODs) and two *Eigenmannia loreitana* (wave-type EODs, forming a beat).

(B) Schematic of logger electrode configuration and fish positions in (A).

(C) Change in knifefish transit rate relative to baseline without eel EODs, estimated with a time-series GLMM (106,920 1-min recordings), with cross-correlation to assess lead-lag structure. Increases at lag 0 (the same minute eel EODs were detected) indicate immediate evasive movements. Positive lags (+1 min, +2 min, etc.) reflect elevated movements in subsequent minutes, and a negative lag (–1 min) indicates the arrival of knifefish displaced by an approaching eel not yet detected by the logger. Points and error bars denote model estimates $\pm$ SE, and p values indicate significant deviations from baseline.

See also Figure S1, Tables S1 and S2, and Video S1.

and stored as consecutive 60-s two-channel recordings on an onboard microcontroller. Across 27 nights, this yielded 106,920 min of recording. The loggers were deployed in open sandbank habitats, where they detected electric eels (*Electrophorus vari*) and four knifefish species—*Gymnorhamphichthys rondoni*, *Eigenmannia loreitana*, *Sternopygus macrurus*, and *Apteronotus albifrons*—two of which (*G. rondoni* and *E. loreitana*) accounted for 92.2% of knifefish detections.²²

Because gymnotiform fishes emit EODs persistently across life stages and sexes, our loggers effectively captured virtually all signals from eels and knifefish passing nearby. Unlike acoustic signals, electric signals are largely unaffected by environmental distortions, preserving fine time-frequency features with exceptional fidelity.²³ This enabled us to reliably distinguish gymnotiform species from their recorded EODs using automated short-time Fourier analysis²² (Figure 1C).

Eel EODs were detected in an average of 14 1-min recordings per logger-night (SD = 10; range = 3–66). Each logger had an omnidirectional detection radius of $\sim$ 8 m for bursts of high-voltage EODs (to $\sim$ 600 V) used in predation and defense²⁴ and $\sim$ 4 m for low-voltage EODs used in electrolocation (to $\sim$ 50 V²⁵). Although low- and high-voltage eel EODs differ in

amplitude, and therefore detection range, they are indistinguishable in waveform shape and spectral composition²⁶ (Figure 1).

Eels were detected only at night, never during the day or civil twilight. Many of the detections likely represent repeated visits by members of a small group of three to four adult eels ($\sim$ 1–1.5 m length) that sheltered by day in a deep pool $\sim$ 200 m upstream of the transect. Recordings with more than one eel were rare (0.8%), consistent with solitary foraging, and we observed no evidence of the group hunting behavior reported for *Electrophorus voltai*.²⁷

Knifefish, which produce weak (<1 V) EODs,^{3,4} were detected in 344 1-min recordings per logger-night (SD = 145; range, 67–730) within a detection radius of $\sim$ 0.3 m, with detections tallied per species per minute and then summed across species.

The high detection rates of both eels and knifefish indicate regular co-occurrence at the logger sites and thus frequent opportunities for predator-prey encounters.

Eel detection and evasion by knifefish

Knifefish, small (mostly <20 cm length) consumers of benthic invertebrates, produce either pulse- or wave-type EODs⁴

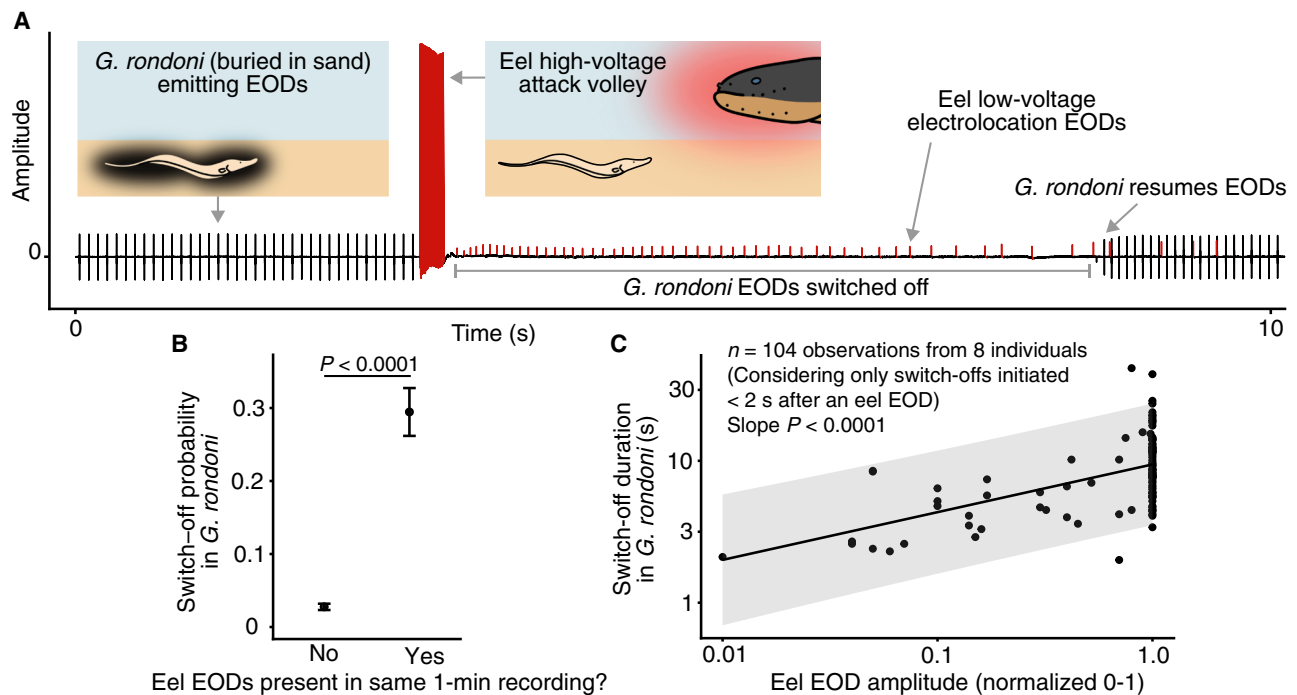


Figure 3. *Gymnorhamphichthys rondoni* switch off their EODs in response to electric eel EODs

(A) Logger recording showing *G. rondoni* (buried in sand) switching off its EOD (black) after a high-voltage eel attack volley (red) and resuming EODs.

(B) Switch-off probability is higher when eel EODs occur in the same 1-min recording (binomial GLMM; 8 *G. rondoni*). Points and error bars denote model estimates $\pm$ SE.

(C) Switch-off duration increases with eel EOD amplitude (GLMM). The shaded region shows 95% confidence intervals with random effects. Clustering at amplitude = 1.0 reflects clipping of eel EODs.

See also Table S3, Video S1, and Data S1.

(Figure 1A). They are preyed upon by electric eels,^{28–32} apex predators that can reach 2.2 m in length.³³ In addition to tuberous electroreceptor-mediated active electrolocation, all gymnotiforms have passive electroreception, in which ampullary electroreceptors detect external low-frequency (<60 Hz) bioelectric cues. In knifefish, ampullary electroreceptors primarily detect ultra-weak (<1 mV) fields emanating from invertebrate prey but are also highly sensitive to the monophasic, low- and high-voltage pulse-type EODs of electric eels, which are dominated by low-frequency energy³² (Figure 1D). Given this sensitivity, we hypothesized that knifefish would exhibit evasive behaviors in response to eel signals.

Consistent with this hypothesis, knifefish movement rates increased markedly during periods in which electric eel EODs were detected by our loggers.

Knifefish maintain stable EOD amplitudes over seconds to minutes, allowing discrete movements (“transits”) to be quantified from fluctuations in EOD amplitude recorded by the logger electrodes.²² Transit rates increased sharply during 1-min windows with low- or high-voltage eel EODs. In *Eigenmannia loreana*, transits rose by 32% ($p < 0.0001$, $z = 9.40$), and in *Gymnorhamphichthys rondoni* by 46% ($p < 0.0001$, $z = 8.80$), relative to eel-free minutes (Figure 2C; Table S1; examples in Video S1A). In both species, transit rates remained elevated for 1–2 min after eel EODs were detected by the logger (Figure 2C; p and z values in Table S2). *G. rondoni* also increased movement in the minute before eel EODs were detected, consistent with

displacement by an approaching eel ($p = 0.011$, $z = 2.55$; Figure 2C; Table S2).

EOD suppression in knifefish

Just as knifefish can detect electric eel EODs via ampullary electroreceptors, eels can detect the electric fields of knifefish.³¹ The conspicuousness of knifefish EODs to eels and other electroreceptive predators likely drove an early evolutionary shift from simple, low-frequency-dominated monophasic EODs to multiphasic EODs with reduced low-frequency content³² (Figure 1A). Nevertheless, many knifefish species have retained some low-frequency EOD components, making them detectable to eels at close range.^{31,32,34–36}

In pulse-type (but not wave-type) knifefish species, individuals sometimes switch off their EODs for seconds to several minutes during courtship or territorial encounters.^{37,38} We hypothesized that they could exploit this same switch-off behavior to avoid detection when encountering electric eels.

Our recordings revealed that *G. rondoni*—the only pulse-type knifefish species detected by our loggers—frequently switched off its EODs when electric eels were present (Figures 3A and 3B).

Gymnorhamphichthys rondoni buries itself in sandbanks during the day and emerges to forage on dark nights, but most individuals remain buried during moonlit periods.²² Buried *G. rondoni* continue to emit EODs (except during switch-offs), permitting night-long monitoring within the detection range of a logger. Among eight such resting *G. rondoni*, switch-offs were

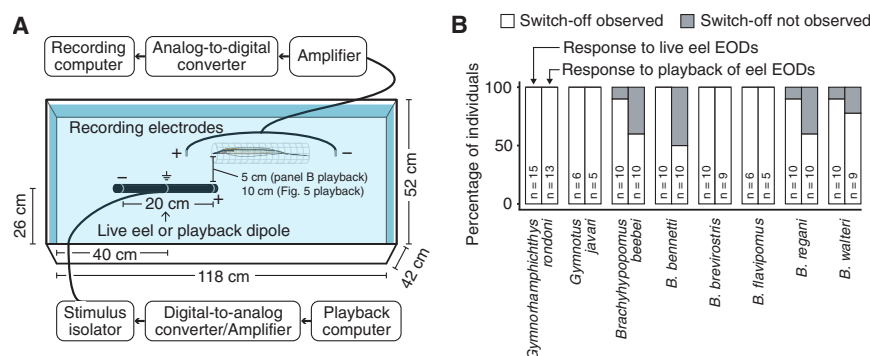


Figure 4. Survey of EOD switch-offs in eight knife-fish species in response to a live electric eel and to playback of eel EODs

(A) Experimental tank and playback setup. In live-eel trials, the playback dipole was replaced by a 56 cm eel.

(B) Percentage of individuals from eight pulse-type knife-fish species with EOD switch-offs in response to a live eel or to playback of low-voltage eel EEs. See also [Video S1](#) and [Data S1](#).

observed in 107 of 351 1-min recordings containing eel EODs (30.5%) ([Video S1B](#)). By contrast, switch-offs occurred in significantly fewer recordings (120 of 4,012, 3.0%) without eel EODs (binomial generalized linear mixed-effect model [GLMM]: $p < 0.0001$, $z = 17.05$; [Figure 3B](#); [Table S3](#)). Switch-off durations ranged from 2 to 41 s and increased with the eel EOD amplitude recorded at the logger (GLMM: $p < 0.0001$, $z = 6.62$; [Figure 3C](#); [Table S3](#)), suggesting that *G. rondoni* assesses threat level from the eel's signal strength, which largely reflects distance from the eel.

In another stream, we recorded eel-induced EOD switch-offs in two *Brachyhyopomus* species inhabiting marginal rootmats ([Data S1](#); [Video S1C](#)). Similar behavior has been documented in *Gymnotus carapo*.²⁸

To assess the prevalence of eel-induced switch-offs among pulse-type knife-fish species, we exposed 77 individuals from 8 upper Amazon species—*Gymnotus javari* (Gymnotidae), *Gymnohamphichthys rondoni* (Rhamphichthyidae), and six *Brachyhyopomus* species (Hypopomidae)—to low- and high-voltage EODs from a live electric eel (56 cm) in controlled laboratory trials ([Figure 4A](#); see [experiment 1](#) in [STAR Methods](#)). EOD switch-offs occurred in all eight species and in 74 of 77 individuals (96.1%) ([Figure 4B](#); [Data S1](#)).

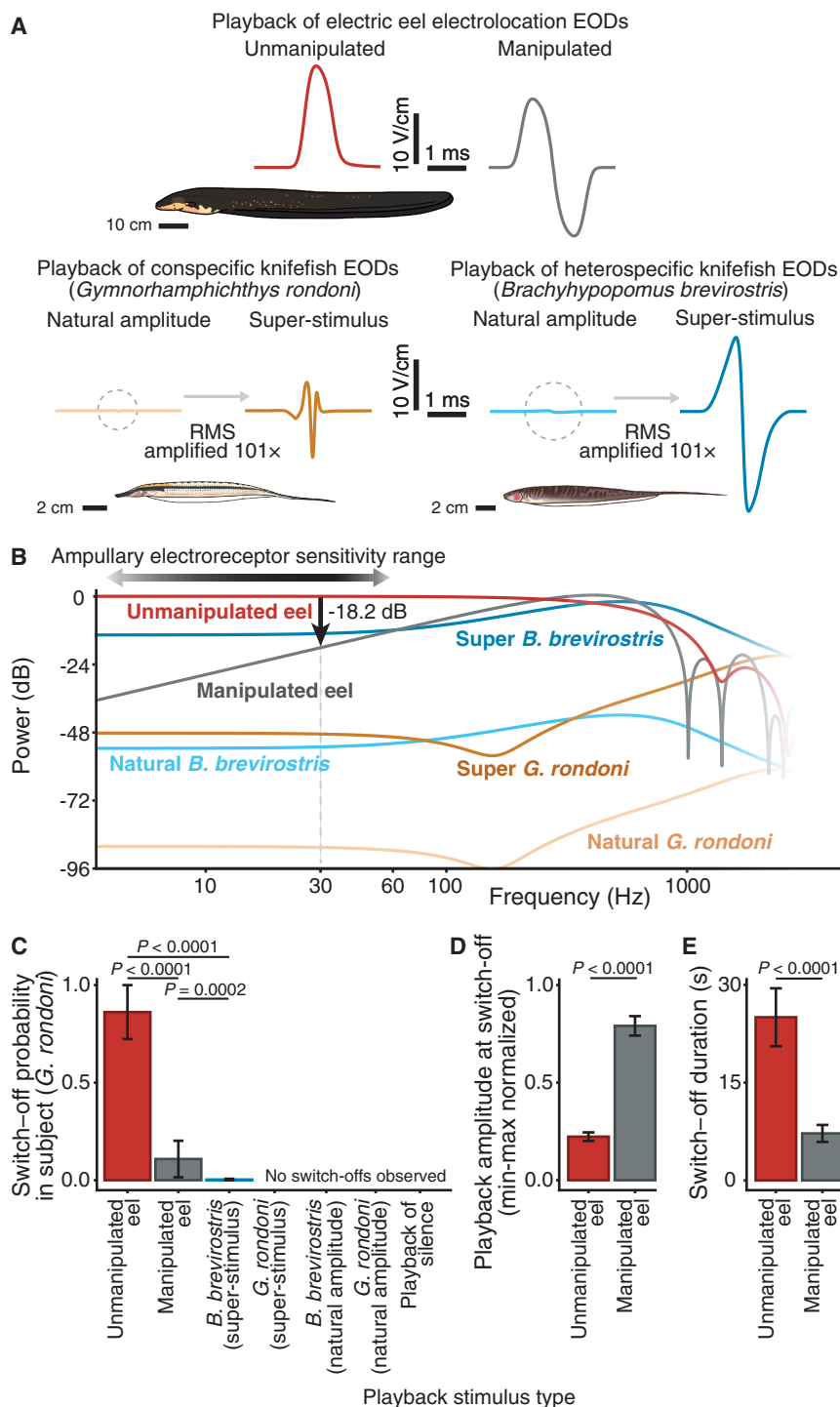
To test whether eel EODs alone trigger switch-offs, absent olfactory, mechanical, or other non-electric cues, we replaced the live eel with playback—delivered through a submerged dipole—of a low-voltage electrolocation EOD sequence recorded from a hunting eel ([Figure 4A](#); see [experiment 2](#) in [STAR Methods](#)). The playback amplitude was ramped up over 30 s to simulate an approach, reaching a realistic maximum peak-to-peak amplitude of 19,200 mV·cm⁻¹—representative of an average-sized eel—measured as the underwater field strength at 55 μ S·cm⁻¹ and 27°C.³⁹ Switch-offs were observed in all *G. rondoni* and *G. javari* and in 50%–100% of individuals across the six *Brachyhyopomus* species ([Figure 4B](#); [Data S1](#)).

To test whether the low-frequency EOD energy (LFE) sensed by knife-fish ampullary electroreceptors contributes to the switch-off response, we conducted controlled laboratory playback experiments in which we selectively reduced LFE in eel waveforms ([Figures 5A–5E](#)). Gymnotiform ampullary electroreceptors are most sensitive at 30 Hz.³² To reduce LFE, we appended a symmetrical negative phase to each monophasic eel EOD, generating biphasic waveforms that matched the root mean square (RMS) amplitude of the originals ([Figure 5A](#)) but

had 18.2 dB less energy at 30 Hz (64.8 $\times$ less power; [Figure 5B](#)) (see [experiment 3](#) in [STAR Methods](#)). All other playback parameters were held constant. Using *G. rondoni* as a test species because of its frequent switch-offs in the field, we found that unmanipulated eel EODs triggered significantly more switch-offs (86.2% $\pm$ 14.8% [mean $\pm$ SE] of trials) than EODs manipulated to reduce LFE (10.9% $\pm$ 9.4%) (binomial GLMM with randomized repeated-measures design: $p < 0.0001$, $z = 5.98$; [Figure 5C](#); [Data S1](#); [Table S4](#); [Video S1D](#)). Unmanipulated eel EODs also triggered switch-offs at lower playback amplitudes (GLMM: $p < 0.0001$, $z = -12.84$; [Figure 5D](#); [Table S5](#)) and induced longer switch-off durations (GLMM: $p < 0.0001$, $z = 7.01$; [Figure 5E](#); [Table S5](#)) than manipulated EODs. Together, these results indicate that LFE plays a key role in triggering EOD suppression, with unmanipulated eel EODs eliciting more frequent, longer, and faster-onset suppression than LFE-reduced stimuli.

To test whether *G. rondoni* switch off their EODs in response to naturally encountered non-eel signals, we played back conspecific EODs and heterospecific EODs from the syntopic knife-fish species *Brachyhyopomus brevirostris*. The playbacks were presented at realistic natural amplitudes for large adults (130 mV·cm⁻¹ for *G. rondoni* and 388 mV·cm⁻¹ for *B. brevirostris*, at 55 μ S·cm⁻¹ and 27°C) and at realistic nighttime EOD rates, using the same amplitude ramp described earlier. No switch-offs were observed in response to either signal type. Even when presented as super-stimuli—with RMS amplitude increased to 101 $\times$ natural levels—conspecific signals failed to elicit switch-offs, and *B. brevirostris* signals elicited very few (0.3% $\pm$ 0.4%) ([Figure 5C](#); [Data S1](#); [Table S4](#)). Notably, although the *B. brevirostris* super-stimulus EODs were presented at the same RMS amplitude as manipulated eel EODs and contained 4.9 dB more energy at 30 Hz (3.2 $\times$ more power) ([Figures 5A](#) and [5B](#)), they provoked significantly fewer switch-offs ($p = 0.0002$, $z = -3.71$; [Figure 5C](#); [Table S4](#)). This indicates that, although LFE contributes to triggering switch-offs, additional parameters—such as EOD rate or waveform structure—must also play a role. Waveform structure is intrinsically linked to LFE, because long and especially asymmetrical waveforms contain more low-frequency energy.

These findings demonstrate that EOD suppression occurs in all three pulse-type knife-fish families. How knife-fish distinguish eel EODs from non-threatening signals, however, remains only partially resolved.



Electrolocation EOD suppression in electric eels

Eels emit low-voltage EODs for electrolocation at low, intermittent rates (Figure 1D), with discharge frequency rising to ~10–30 Hz during activity.^{40,41} They also produce high-voltage EOD bursts that begin at ~200–500 Hz^{40,42} (Figure 1D; Video S1E), activating motor neurons and inducing muscle contractions in nearby animals.⁴³ Although these electric discharge behaviors are well documented in captivity,^{40–45} little is known about the electrogenic

Figure 5. EOD switch-offs in *Gymnorhamphichthys rondoni* in response to playback of electric eel and knifefish signals

(A) Six playback stimuli, shown at maximum playback amplitude. Electric field strength and time scales apply to all six waveforms. (B) Power spectra of stimuli in (A), highlighting energy near 30 Hz.

(C–E) GLMMs showing effects of stimulus type on switch-off probability (C), playback amplitude at switch-off (D), and switch-off duration (E) in *G. rondoni* (7 repeats per individual; 13 individuals). Error bars denote model estimates $\pm$ SE. See also Tables S4–S6, Video S1, and Data S1.

behavior of wild electric eels. Because their signals are readily detected by knifefish, we hypothesized that hunting eels would modulate electrolocation EODs in an on-off manner, pausing emission during predatory approaches to reduce detectability.

Our findings support the hypothesis that wild electric eels temporarily silence electrolocation EODs during predatory approaches, resuming electrolocation only after emitting high-voltage bursts that inevitably reveal their presence to electroreceptive prey.

We restricted analyses to “newly initiated bursts,” defined as those not preceded by another burst for ≥ 30 s, which we interpreted as discrete hunting events near a logger. This criterion focused our analyses on encounters in which prey were unlikely to have been recently exposed to predatory eel activity—conditions under which maintaining silence would offer the greatest stealth advantage. To ensure that concurrent low-voltage EODs, if present, would have been detectable, we included only newly initiated bursts with peak amplitudes ≥ 0.85 on the logger’s 0–1 normalized scale, a threshold at which low-voltage EODs are detectable by the logger within ~4 m based on range calibration tests with an average-sized eel (see [detection range calibration](#) in STAR Methods).

Inferring suppression required distinguishing true silences from false negatives, in which electrolocation EODs were undetected because the eel was outside the logger detection range.

To minimize false negatives, we searched for electrolocation EODs in 15-s windows before and after each newly initiated burst. This window length, derived from measured eel swim speeds and logger detection ranges, balanced the maximum plausible duration of a detectable silence against confidence in distinguishing true from false silence.

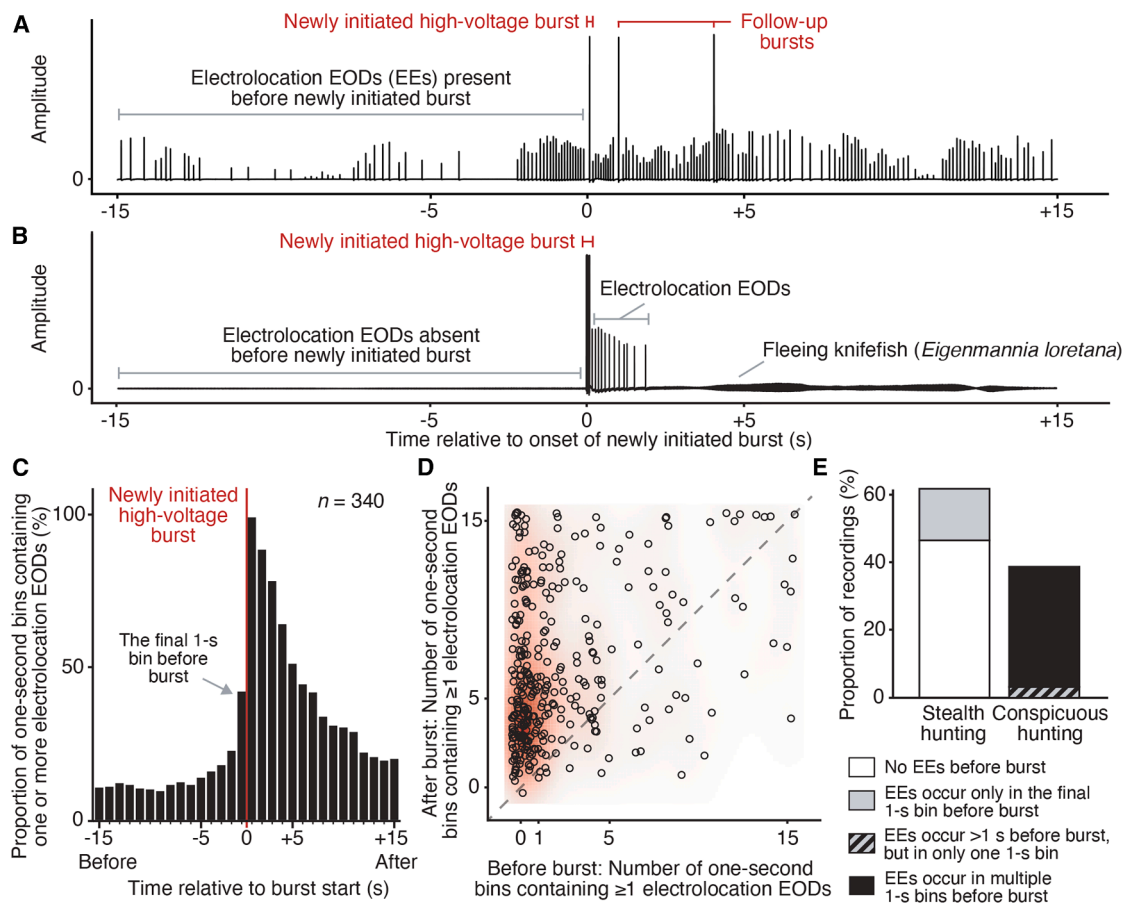


Figure 6. Stealth and conspicuous electrolocation behavior in electric eels

(A) Segment of a conspicuous hunting sequence.

(B) Segment of a stealth hunting sequence with 15 s of electric silence (no electrolocation EODs [EEs]) before a newly initiated high-voltage burst, followed by a train of EEs. Bursts are clipped in (A) and (B).

(C) Proportion of 1-s bins containing ≥ 1 EEs in the 15 s before and after a newly initiated burst (burst not preceded by another for ≥ 30 s) (averaged across 340 newly initiated bursts).

(D) Number of 1-s bins containing ≥ 1 EE in the 15 s before versus after each newly initiated burst. Points ($n = 340$) are jittered to reduce overlap and shaded by density. The diagonal line indicates parity between pre- and post-burst electrolocation activity.

(E) Classification of hunting events based on the distribution of EEs in the 15 s preceding newly initiated bursts ($n = 340$).

See also Video S1.

Across 340 newly initiated bursts meeting the ≥ 0.85 threshold criterion, 61% were classified as “stealth hunting,” defined as events with no low-voltage electrolocation EODs during the full 15 s preceding the burst (46% of all events) or with EODs only in the final second (15% of events) (Figures 6B–6E; Video S1F). In nearly all stealth hunting episodes, eels resumed low-voltage EODs immediately after the burst, often as a rapid sequence lasting >5 s (Figures 6B–6D). The remaining 39% of hunting events were classified as “conspicuous hunting,” with electrolocation EODs occurring 1–15 s before the burst and typically continuing afterward (Figures 6A and 6E).

Our approach likely underestimated silence duration because our analysis window extended only 15 s before each burst, truncating any silences that began earlier. A natural history observation from a stream in French Guiana documented an eel remaining silent for 55 s before emitting a high-voltage discharge and capturing a knifefish

(*Gymnotus carapo*).²⁸ This remains the only previously published description of the electric-signaling behavior of a wild electric eel.

High-voltage predatory discharges in electric eels

Eels are known to produce two types of high-voltage discharges (Figures 7A–7C): (1) “attack volleys,” prolonged bursts that immobilize prey,^{42–44} facilitate electrolocation of prey position,⁴⁶ and serve in defense,⁴⁵ and (2) “twitch inducers,” short bursts of similar amplitude that probe for hidden prey by eliciting muscular spasms that generate water disturbances detectable to the eel’s mechanosensory lateral line system.^{42,43} Both high-voltage burst types broadcast an electric eel’s presence widely to electroreceptive prey. We therefore hypothesized that eels would deploy bursts sparingly to balance predatory effectiveness with stealth, potentially also reducing energetic costs.

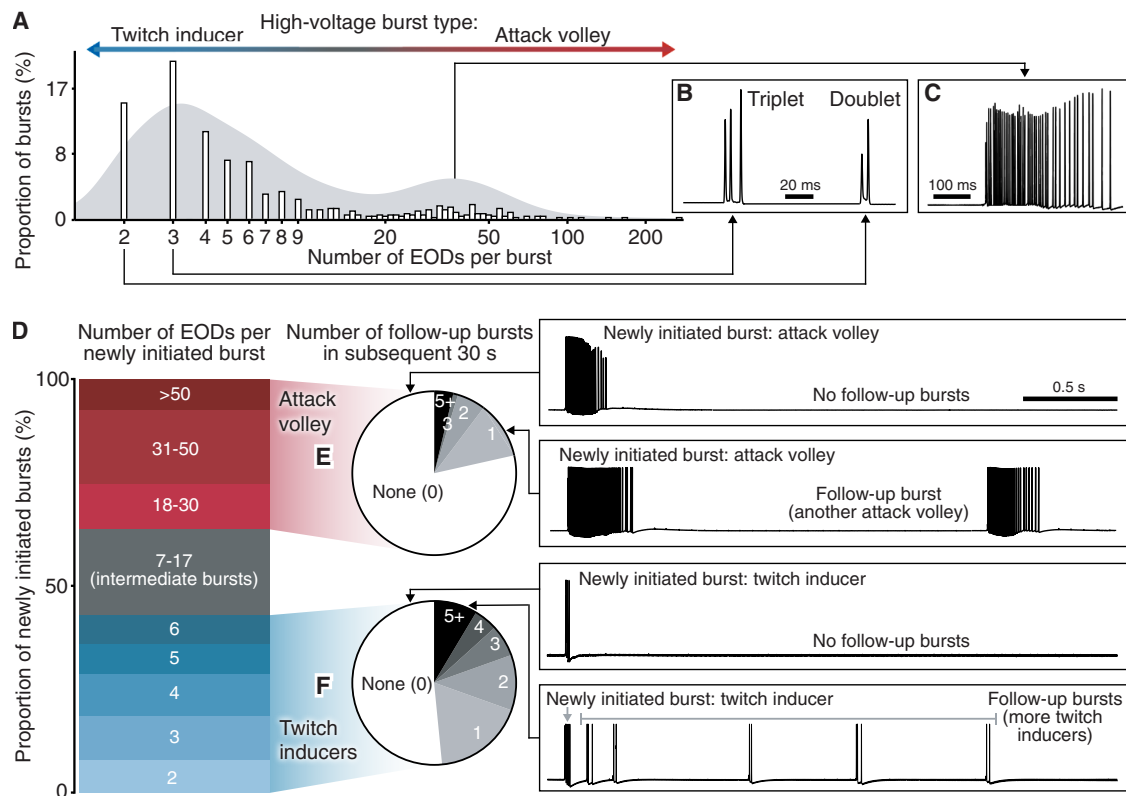


Figure 7. Structure and deployment of high-voltage bursts in electric eels

(A–C) (A) Frequency distribution of EODs per burst across 593 bursts (newly initiated and follow-up combined), with a smoothed density estimate (gray). Bursts are categorized as twitch inducers (2–6 EODs; examples in B), attack volleys (≥ 18 EODs; example in C), or intermediate bursts (7–17 EODs). (D–F) (D) Composition of 296 newly initiated bursts (bursts not preceded by another for ≥ 30 s), grouped by EOD count and classified as twitch inducers (blue), intermediate (gray), or attack volleys (red). Data in (D)–(F) represent a subset of the 340 newly initiated bursts analyzed in Figures 6C–6E, excluding 44 recordings with strong signal clipping. (E and F) Number of follow-up bursts (either twitch inducers or attack volleys) within 30 s of an initial attack volley (E) or twitch inducer (F). Follow-up bursts after intermediate bursts are not shown (64% have none). Right-hand panels show representative logger recordings: two attack volleys (one with and one without follow-up bursts) and two twitch inducers (one with and one without follow-up bursts). Baseline deviations during bursts are artifacts of transient capacitor overload in the logger amplifier.

See also Video S1.

Consistent with this hypothesis, high-voltage discharges in our field recordings were typically isolated in time, rather than occurring in rapid succession.

Our loggers captured 593 high-voltage bursts, containing 2–266 EODs, with a bimodal distribution in burst length (Figure 7A). In captive studies, twitch inducers occur exclusively as doublets or triplets.^{42,43} In our recordings, doublets and triplets were again the dominant twitch-inducer types, comprising 15% and 20% of bursts, respectively (Figures 7A and 7B). We also observed bursts containing 4–6 EODs, which we interpret as serving the same twitch-inducing function. We classified bursts with 2–6 EODs as twitch inducers and those with ≥ 18 as attack volleys (Figures 7A and 7C). Bursts of intermediate length (7–17 EODs) may represent curtailed attack volleys or a hybrid of probing and attack behavior.

Considering only newly initiated high-voltage bursts ($n = 296$), treated as separate hunting episodes, 43% were twitch inducers, 36% were attack volleys, and 21% were of intermediate length (Figure 7D). Among twitch inducers, 51% were isolated—not followed by additional bursts within 30 s (labeled

“none” in Figure 7F)—consistent with a single twitch inducer being sufficient to confirm the absence of nearby prey. Attack volleys often occurred without preceding twitch inducers, indicating that eels frequently attack prey without probing first (Figure 7D). Moreover, attack volleys were isolated even more frequently than twitch inducers, with 79% not followed by another burst within 30 s (Figure 7E; Video S1G).

DISCUSSION

Our results indicate that both electric eels and their knifefish prey engage in context-dependent electric stealth, alternating between generating and silencing their electrolocation EODs to balance active-sensing performance against the risk of mutual detection. Knifefish also increased movement during periods when eel EODs were detected, consistent with evasive responses.

A limited form of electric stealth has been documented in cuttlefish and embryonic elasmobranchs that reduce ventilatory movements when approached by electroreceptive elasmobranch

predators, thereby suppressing weak (<1 mV) bioelectric fields generated as by-products of ventilation.^{47,48} However, this prey-only response differs fundamentally from the electric stealth we report in gymnotiforms, which possess specialized electric organs that generate much larger, higher-amplitude fields for active sensing.⁴⁹

Electric eels frequently suppress low-voltage electrolocation EODs while hunting, breaking their electric silence only to emit the high-voltage EOD bursts used to probe for or immobilize prey. Following such a burst—which inevitably announces their presence to electroreceptive prey—eels promptly resume electrolocation. In this post-burst phase, stealth is no longer advantageous. Instead, active electrolocation becomes critical for close-range localization and capture of prey, including targets incapacitated by an attack volley or revealed by twitch-inducer-evoked water displacements.

Knifefish, for their part, frequently respond to eel EODs by pausing their own EODs, contributing to a two-sided pattern of silence between predator and prey. Switch-offs were observed in all three pulse-type knifefish families, indicating that the response is phylogenetically widespread and likely represents a common adaptive strategy against electric eel predation.

Knifefish also responded to eel EODs with evasive movement. On average, a given logger detected eel EODs ~ 14 times per night, indicating repeated exposure of knifefish to eel signals. These recurring disturbances create a “landscape of fear”⁵⁰ in which prey activity and signaling behavior are shaped by electric eel EOD cues alone, even in the absence of direct predation. Analogous effects have been described in other sensory modalities,^{21,51–53} but our observations provide the first evidence for an electrosensory landscape of fear.

Modern *Electrophorus* species diverged 3–14 million years ago,²⁴ and the lineage itself separated from weakly electric gymnotiforms some 50–70 million years ago,^{54,55} suggesting that the stealth interactions and fear-driven behaviors revealed here may have been shaped over tens of millions of years of predator-prey coevolution.

In a broader comparative context, the stealth dynamics we document here between electric eels and knifefish represent a compelling case of evolutionary convergence with acoustic interactions between killer whales and their echolocating odontocete prey. In both systems, predators suppress active-sensing signals when stalking prey, while prey respond by fleeing or silencing their own signals.^{15–20} Silences in the electrosensory and auditory domains suspend the primary active sense of both predator and prey, limiting their ability to detect and track objects.

This convergence reflects a shared constraint in predator-prey systems where both parties rely on—and can intercept—signal-emission-based active sensing. In such cases, silence reduces detectability but also forfeits the information required for spatial localization. This constraint favors regulated on-off modulation rather than sustained silence throughout an interaction. This contrasts with predator-prey interactions involving communication signals, where extended silence carries relatively little functional cost.

During silent periods, eels and knifefish likely rely in part on their mechanosensory lateral line to sense water movements and avoid obstacles,^{43,56} a strategy shared widely by fish

inhabiting low-light environments,⁵⁷ and on their ampullary electroreceptors to passively detect low-frequency fields from nearby prey.⁵⁸ Comparable shifts in sensory modality occur in the killer whale-odontocete system, where both predator and prey fall back on passive acoustic cues when active signaling is suppressed.^{15,16} Most odontocetes operate in open water, where suppressing echolocation entails minimal risk of disorientation or collision.¹⁵ Electric eels are similar in this respect: they move slowly, face a low risk of injury from collisions, and may partly compensate for interruptions in electrolocation through passive senses.

The bidirectional predator-prey signal-suppression dynamic shared by the eel-knifefish and killer whale-odontocete systems has not, to our knowledge, been reported in other animal groups. A natural question is why a comparable suppression pattern has not evolved in bats (the other major group of echolocating animals) or in mormyrids (the other major group of electrolocating fishes³). The simplest explanation is ecological: predator-prey interactions in which both predator and prey rely on signal-emission-based active sensing are rare or absent in these groups. Among bats, such interactions are rarely documented, with the possible exception of opportunistic predation by spectral bats on other bat species.⁵⁹ Even so, although some bats reduce call amplitude dynamically to lower detectability to eared moth prey⁶⁰ (we observed no evidence of analogous EOD amplitude modulation in gymnotiforms), complete cessation of echolocation is rare. This likely reflects the need for near-continuous acoustic feedback to avoid collisions during high-speed navigation in cluttered environments.^{8,12–14}

Beyond biological systems, signal-suppression dynamics comparable to those observed in the electric eel-knifefish and killer whale-odontocete-prey systems are also found in human active-sensing technologies. For example, sonar and radar operators on World War II warships and submarines continually weighed the need to detect the enemy against the risk of revealing their own presence and therefore alternated strategically between active operation and passive “silent running.”^{61,62}

Our findings identify several directions for future work. On the predator side, it remains unclear how habitat complexity influences transitions between low-voltage electrolocation EOD emission and EOD cessation in electric eels. For example, movement through cluttered microhabitats may increase the need for continuous spatial sampling. It also remains unclear whether these transitions depend on prey type—in particular, electroreceptive knifefish or catfish versus non-electroreceptive animals—because the benefits of silence should differ sharply depending on whether prey can detect eel EODs. On the prey side, the neural mechanisms underlying context-dependent EOD switch-offs in knifefish remain unresolved. Signal suppression may be mediated by neuromodulatory control of the medullary pacemaker nucleus, which regulates EOD rhythm during social interactions.⁶³ Determining how sensory input from ampullary electroreceptors is integrated with this central pacemaker circuitry will clarify how external threat cues are translated into rapid, reversible suppression of EOD production. A further question is whether knifefish species or populations that have experienced prolonged allopatry from electric eels show reduced or absent switch-off responses to eel EODs, consistent with relaxed selection on this defensive behavior.

This study reveals electric stealth between eels and knifefish as a key predator-prey interaction in Amazonian aquatic systems, reflecting an evolutionary arms race of detection and concealment with ancient roots. It also provides the first detailed account of electric-signaling behavior in wild electric eels, expanding our understanding of the Amazon's nocturnal bioelectric environment. More broadly, our findings show that in systems relying on signal-emission-based active sensing, predators and prey alike face a fundamental trade-off: signals that are essential for acquiring spatial information are also inherently conspicuous. Across electrosensory, acoustic, and engineered systems, this conflict has repeatedly been resolved through intermittent, on-off modulation of active sensing.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to the lead contact, William G.R. Crampton (crampton@ucf.edu).

Materials availability

This study did not generate any new, unique reagents.

Data and code availability

- All data used in this paper have been deposited in Dryad: <https://doi.org/10.5061/dryad.1ns1m95w> and are publicly available as of the date of publication.
- All original code has been deposited in Dryad: <https://doi.org/10.5061/dryad.1ns1m95w> and is publicly available as of the date of publication.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Both authors conceived and designed field procedures and laboratory experiments, performed fieldwork, conceptualized the analytical approaches, and wrote the manuscript. L.P. performed the playback experiments, executed the statistical analyses, and produced the illustrations. W.G.R.C. designed the EOD loggers and other electronic apparatus and associated software.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Raw and analyzed data as well as R code used	This paper	Dryad: https://doi.org/10.5061/dryad.1ns1rn95w
Experimental models: Organisms/strains		
<i>Electrophorus vari</i>	Observed in and collected from the wild	N/A
<i>Gymnorhamphichthys rondoni</i>	Observed in and collected from the wild	N/A
<i>Gymnotus javari</i>	Collected from the wild	N/A
<i>Brachyhypopomus beebei</i>	Collected from the wild	N/A
<i>Brachyhypopomus bennetti</i>	Collected from the wild	N/A
<i>Brachyhypopomus brevirostris</i>	Collected from the wild	N/A
<i>Brachyhypopomus flavipomus</i>	Collected from the wild	N/A
<i>Brachyhypopomus regani</i>	Collected from the wild	N/A
<i>Brachyhypopomus walteri</i>	Collected from the wild	N/A
<i>Brachyhypopomus sullivani</i>	Observed in the wild	N/A
<i>Brachyhypopomus verdii</i>	Observed in the wild	N/A
<i>Eigenmannia loreana</i>	Observed in the wild	N/A
Other		
Custom-designed EOD logger	Poon et al. ²²	Zenodo: https://doi.org/10.5281/zenodo.15073474
Custom-designed stimulus isolator	This paper	N/A
Custom-designed playback dipole	This paper	N/A
Playback digital-to-analog converter with amplifier	RME Audio, Haimhausen, Germany	ADI-2 DAC FS
Recording analog-to-digital converter	National Instruments, Austin, TX, USA	USB-6216-BNC
Bioamplifier	Ametek Scientific Instruments, Oak Ridge, TN, USA	SR-5113
Bioamplifier	CWE Inc., Ardmore, PA, USA	BMA-200
Tablet personal computer	Dell, Round Rock, TX, USA	Rugged Extreme 7220
Audacity	Muse Group	https://www.audacityteam.org/
Decibel	Feisty Dog, LLC	https://www.feisty-dog.com/decibel/
LabVIEW	National Instruments, Austin, TX, USA	https://www.ni.com/en/shop/labview.html

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

All laboratory experiments and field procedures were approved by the UCF Institutional Animal Care and Use Committee (IACUC), under protocol numbers 07-45, 17-24W, and 202300073.

Subjects used in Experiment 1

Knifefish used in “Experiment 1” (“EOD switch-offs of eight knifefish species in response to a live eel”) were collected by licensed ornamental fish catchers near Iquitos, Peru.

Five species were collected in rainforest streams of the Río Nanay and Río Itaya:

Rhamphichthyidae: *Gymnorhamphichthys rondoni* (sand knifefish) (n = 15; 130–172 mm total length [TL]).

Gymnotidae: *Gymnotus javari* (tiger banded knifefish) (n = 6; 103–183 mm TL).

Hypopomidae: *Brachyhypopomus beebei* (n = 10; 97–174 mm TL) (bluntnose knifefish); *B. brevirostris* (bluntnose knifefish) (n = 10; 105–270 mm TL); *B. walteri* (bluntnose knifefish) (n = 10; 110–163 mm TL).

Three species were collected in floating macrophytes along the margins of whitewater floodplain lakes of the Río Amazonas: Hypopomidae: *Brachyhypopomus bennetti* (bluntnose knifefish) ($n = 10$; 90–130 mm TL); *B. flavipomus* (bluntnose knifefish) ($n = 6$; 62–70 mm TL); *B. regani* (bluntnose knifefish) ($n = 10$; 80–102 mm TL). Multiple *Brachyhypopomus* species were included because the genus is diverse and abundant across habitats and shows wide variation in EOD waveform and rate.⁶⁴ All individuals were healthy and above minimum reproductive size.^{65–67} Each species was represented by both males and females, but none were in breeding condition (gonads undeveloped, stages 1–2, following Brown-Peterson et al.⁶⁸ and Waddell and Crampton⁶⁹). Outside playback trials, knifefish were housed in sand-bottom aquaria with shelters and air-driven sponge filtration and were fed live *Tubifex* worms or blackworms (*Lumbriculus* sp.) *ad libitum*.

A single 56 cm electric eel (*Electrophorus vari*), used to elicit knifefish EOD switch-offs, was collected in a rainforest stream near Iquitos by ornamental fish catchers and was fed a variety of small live fish (characiforms and gymnotiform knifefish) outside trials.

Subjects used in Experiment 2

The same eight knifefish species (but not the same individuals) from Experiment 1 were used in “Experiment 2” (“EOD switch-offs of eight knifefish species in response to playback of eel electrolocation EODs”). Fish were collected by ornamental fish catchers from the same habitats near Iquitos, Peru. Sample sizes and body size ranges matched those of Experiment 1 as closely as possible: *Gymnorhamphichthys rondoni* ($n = 13$; 158–176 mm TL); *Gymnotus javari* ($n = 5$; 123–138 mm TL); *Brachyhypopomus beebei* ($n = 10$; 116–156 mm TL); *B. bennetti* ($n = 10$; 94–122 mm TL); *B. brevirostris* ($n = 9$; 101–185 mm TL); *B. flavipomus* ($n = 5$; 65–73 mm TL); *B. regani* ($n = 10$; 82–99 mm TL), and *B. walteri* ($n = 9$; 124–167 mm TL). Fish were collected 6–8 weeks before import to our laboratory through a licensed ornamental fish export company.

Subjects used in Experiment 3

The 13 *Gymnorhamphichthys rondoni* used in Experiment 2 were also used in “Experiment 3” (“EOD switch-off responses of *G. rondoni* to seven playback stimuli”). Knifefish used in Experiments 2 and 3 were maintained before and between trials under the same housing conditions described for Experiment 1.

METHOD DETAILS

Field site

Fieldwork was conducted in intact rainforest at the Tanimboca Reserve near Leticia, Colombia (4.128°S, 69.953°W) during July–August 2022. This period corresponds to the regional dry season, when stream level, temperature, and other physico-chemical variables were stable,²² ensuring consistent recording conditions. Stream-inhabiting gymnotiforms are non-reproductive during the dry season,⁷⁰ allowing observations to focus on foraging and predator–prey behaviors.

EOD logger deployment and design

Six EOD loggers were deployed nightly for 27 consecutive nights (July 28–29 to August 23–24), operating from late afternoon to early morning. Analyses were restricted to 18:15–05:15 (11 h), when electric fish are active, thereby excluding civil twilight and the first half of nautical twilight. Each logger recorded continuously, saving consecutive one-minute, two-channel .wav files (one channel per electrode pair) to microSD storage.

The design and operation of the EOD logger and electrode array follow Poon et al.,²² with full schematics and code available at Zenodo.⁷¹ Each logger housed two bioamplifiers and a Teensy 3.5 microcontroller (PJRC, Sherwood, OR) inside a waterproof enclosure on the stream bank, connected via a shielded cable buried beneath the sand to an electrode array anchored midstream. The array comprised two nickel-chromium electrode pairs spaced 60 cm apart and projecting 3 cm into the water column, with one pair aligned parallel to the flow and the other orthogonal to it. Loggers were spaced 15–20 m apart along the 150 m transect to prevent overlap of detection ranges (Figure S1). Signals were band-pass filtered (1.6 Hz–39.9 kHz), amplified (90 $\times$), and digitized at 49,152 Hz and 12-bit resolution using Arduino-based firmware modified from <https://github.com/muchaste/EOD-logger>.²² Because logger-recorded electric field amplitude depends on distance, orientation, body size, and water conductivity, eel EOD amplitudes are expressed on a normalized 0–1 scale rather than as absolute field strengths.

Detection range calibration

To establish the signal thresholds used for automated EOD detection (see “detection of EODs”), we quantified how detectability varied with distance from the logger for *Electrophorus vari* and for the four open-water knifefish encountered in the logger recordings (*Gymnorhamphichthys rondoni*, *Eigenmannia loreana*, *Sternopygus macrurus*, and *Apteronotus albifrons*).

For electric eels, range calibration trials used an average-sized individual (1.1 m TL) captured from the study stream after the logger surveys. The eel was held in a straightened posture inside a plastic mesh tube positioned midstream and parallel to the flow. Low- and high-voltage EODs, elicited by gentle prodding with a plastic dowel, were recorded at 1-m intervals from 1 to 15 m upstream of the eel (Figure S2A). Low-voltage EODs became reliably detectable in both raw time–voltage waveforms and spectrograms when the eel was ~ 4 m from the logger—a distance at which high-voltage EOD peak-to-peak amplitude measured 0.85 on the logger’s normalized 0–1 scale. This relationship between high- and low-voltage EODs at 4 m defined the detectability threshold applied in analyses of eel electrolocation-signal suppression (see “electrolocation EOD suppression in electric eels” in “results”).

For knifefish, analogous tests were conducted by placing average-sized adults in mesh tubes at varying distances, orientations, and angles relative to the logger's orthogonal electrode pairs (Figure S2B). Weak (<1 V) EODs of average-sized adults were detectable within ~30 cm of the electrode-pair centers at the thresholds defined in "detection of EODs"; these empirical results informed the species-specific parameters applied in the automated EOD detection analyses.

Detection of EODs

EODs from eels and knifefish were identified using custom R script ¹⁷² (in R v4.4.2; R Core Team 2024), incorporating functions from the Seewave package.⁷³ The two channels of each one-minute recording were analyzed independently. After removal of DC-offset, we generated spectrograms via a short-time Fourier transform (STFT; 20 ms window, 17.45% overlap), yielding 16.5 ms temporal and 50 Hz frequency resolution (Figure 1C). Frequencies above 8 kHz were excluded from further analysis.

Electric eels

Electric eel EODs were scored as present when two criteria were simultaneously satisfied. First, spectral power had to exceed thresholds of 45, 38, and 31 dB in three low-frequency bands (0–50, 50–100, and 100–150 Hz, respectively). This multi-band criterion prevented misclassification of low-frequency EOD components produced by the wave-type knifefish *Sternopygus macrurus*. Second, to minimize false positives arising from temporal smearing in the STFT, at least three consecutive 16.5 ms bins had to exceed these thresholds. A one-minute recording was classified as containing eel activity (a "presence") if either channel met these detection criteria. The decibel thresholds were derived from empirical range tests (see "detection range calibration") and correspond to signals recorded from an average-sized eel (~1.1 m), detected at ~4 m for low-voltage EODs and ~8 m for high-voltage bursts.

Knifefish

Knifefish EODs were detected using the same STFT-based procedure as for eels, but with an additional temporal-smoothing step to improve signal-to-noise ratio, reflecting the dense temporal structure of their EOD emission. A moving average was applied across 120 successive 16.5 ms bins (50% overlap), yielding one-second analysis segments. Species-specific thresholds were then applied. *Gymnorhamphichthys rondoni* EODs were scored when ≥28 of 120 frequency bands (50 Hz bands between 1,500 and 7,500 Hz) exceeded 12 dB. *Eigenmannia lorentana* EODs were scored when ≥2 of 5 frequency bands (50 Hz bands between 350 and 600 Hz) exceeded 28 dB. Detection thresholds for *Sternopygus macrurus* and *Apteronotus albifrons* followed Poon et al.²²

This procedure yielded one-second-resolution presence-absence time series for each species. All decibel thresholds correspond to signals from average-sized adults detected at ~30 cm from the center of an electrode-pair, as determined by the range-calibration trials.

Knifefish transit rate

Using the knifefish presence-absence time series, we used R script ²⁷² to derive a movement metric, the knifefish transit rate. This analysis included only the two most abundant species, *Eigenmannia lorentana* and *Gymnorhamphichthys rondoni*. Transit rate quantifies the frequency of discrete movements by individuals relative to the EOD loggers.

A transit was defined as a single transition between presence and absence. For example, a fish entering the 30 cm logger detection radius and remaining for one minute would generate 60 consecutive one-second presences but only one transit. Rapid changes in orientation (e.g., U-turns) that produced brief presence-absence switches were also counted as transits. To prevent overcounting repeated movements by the same individual in quick succession, each one-minute interval was subdivided into three 20-s blocks, with at most one transit permitted per block. Transit rate was then calculated as the number of transits per minute. The schematic in Figure S3 summarizes the logic for converting presence-absence time series into transits. Transits detected simultaneously on both logger channels were counted only once.

Detection of switch-offs in wild *G. rondoni*

Using Audacity, we manually searched recordings on moonlit nights for resting *Gymnorhamphichthys rondoni*. Resting individuals were identified by their stable amplitudes (indicating no movement) and by lower pulse rates (~20–40 Hz) compared with active individuals (~70–120 Hz).²² Eight individuals remained buried within logger range for 3–11 h. For each, we recorded the occurrence and duration of EOD switch-offs and noted the presence or absence of eel EODs within the same one-minute recording. For switch-offs occurring within 2 s of an eel EOD, we also measured the eel's EOD amplitude, expressed as a normalized value between 0 and 1 (with 1 indicating signal clipping).

Recordings of switch-offs in *Brachyhyopomus*

EOD switch-offs were observed incidentally in three *Brachyhyopomus* individuals (Data S1) during a 2018 field survey of gymnotiform spatial distributions in rainforest streams near Jenaro Herrera, Peru (04°54'S, 073°40'W). Fish were located one-by-one with an electric fish finder⁷⁴ under dim red light while wading upstream and were recorded with a hand-held EOD recording system. This system comprised two orthogonal electrode pairs (each with 15 cm dipole separation) mounted on a rod and connected to two BMA-200 bioamplifiers, a USB-6216 analog-to-digital converter, and a Dell 7220 Rugged Extreme tablet running custom LabVIEW recording software. Two-channel (stereo) .wav files were recorded for 60 s at 50 kHz and 16-bit resolution. Across the survey, *Electrophorus vari*i were detected in 10 of 1,125 knifefish recordings (0.89%), and the three *Brachyhyopomus* EOD switch-offs occurred in three of those 10 recordings (Data S1).

Classification of low- and high-voltage EODs

The eel's low-voltage electrolocation EODs were distinguished from high-voltage EODs based on repetition rate: low-voltage EODs are invariably <30 Hz, whereas high-voltage EOD bursts exceed 30 Hz.⁴⁰ Because rate analysis requires high temporal resolution, we analyzed the original time–voltage signal rather than spectrograms. Using R script 3,⁷² we rectified the two channels of each file containing eel EODs and averaged them into a mono signal. We then applied a peak-detection routine to timestamp each EOD pulse and calculate instantaneous rates (1/inter-pulse interval).

Recordings with instantaneous rates >30 Hz were manually inspected to verify burst recognition and correct misclassifications. One, or sometimes two, intermediate-voltage EODs of unknown function occur at the onset of each high-voltage burst^{26,40} (example in Figure 1D). These EODs were excluded from counts of both low- and high-voltage EODs. During this process, we also identified recordings containing signals from multiple eels, which were excluded because concurrent EODs from different individuals could not be reliably separated; these recordings were easily recognized by their erratic pulse-rate patterns.

R script 3 was subsequently used to select newly initiated bursts (those occurring ≥ 30 s after any prior burst) from the full set of identified bursts and to retain only events with a peak amplitude ≥ 0.85 on the logger's 0–1 normalized scale (for threshold justification see “[detection range calibration](#)”).

Eel swim speed

To characterize typical movement speeds during hunting, individuals were observed under dim red light along a 25 m stream section before logger recordings began. A grid of thin poles spaced at 2 m intervals provided reference points. Movement was timed with a stopwatch as individuals swam past the grid, and hunting behavior was confirmed by monitoring high-voltage bursts with an electric fish finder.⁷⁴ Eels that traversed the grid without producing twitch inducers (bursts of 2–6 EODs) or attack volleys (bursts of ≥ 18 EODs), or that followed erratic paths, were excluded. Hunting eels swam at $0.12\text{--}0.25\text{ m}\cdot\text{s}^{-1}$ (mean $0.20\text{ m}\cdot\text{s}^{-1}$). These relatively low speeds are consistent with previous reports that eels move slowly before producing bursts, likely to optimize mechanosensory inputs.⁴³

Distinguishing true from false eel silences

The 15 s window before and after newly initiated bursts (see “[electrolocation EOD suppression in electric eels](#)” in results) was selected to balance temporal resolution against detection range: a shorter window risked missing electrolocation activity, whereas a longer window risked including false silences when eels moved beyond the logger's range. With swim speeds of $0.12\text{--}0.25\text{ m}\cdot\text{s}^{-1}$, an eel crossing the logger's detection range in a straight line and producing a burst at the center would remain detectable for 16–33 s, well beyond the 15 s analysis windows.

Nonetheless, eels were expected to produce bursts at arbitrary positions within the logger's detection range. An eel entering the ~ 4 m radius shortly before a burst might have electrolocation EODs missed for part of the pre-burst window but detected throughout the post-burst window; conversely, an eel leaving the radius might be detected before the burst but missed afterward. This symmetry ensured that while some EODs may have gone undetected—particularly from rapidly moving eels—comparisons between pre- and post-burst periods remained unbiased.

Characterization of low-voltage EOD emission

Spectrogram heatmaps were generated by rerunning R script 1 on recordings spanning 15 s before and after each newly initiated burst (examples in Figure 1C and Video S1E). These spectrograms were used to manually score the presence or absence of electrolocation EODs in one-second bins. An electrolocation EOD was identified when power in the 0–150 Hz range exceeded background levels by at least 12 dB. Manual scoring was preferred because individual low-amplitude EODs were readily visible in spectrograms but difficult to detect reliably with automated threshold scripts.

One-second presence–absence bins were used because the temporal distribution of EODs is a more relevant measure of conspicuousness than their total number. For example, an eel emitting 15 EODs within a single one-second bin but remaining silent during the other 14 s would likely be harder for a knifefish to track than one emitting a single EOD every second.

High-voltage EOD burst structure

For all newly initiated bursts, we used the automated peak-detection procedure described under “[classification of low- and high-voltage EODs](#)” to count the number of EODs per burst. Bursts that were heavily clipped, preventing accurate counting, were excluded. The 30 s period following each burst was then examined for follow-up bursts using an automated procedure in R script 4.⁷²

Experiment 1

EOD switch-offs of eight knifefish species in response to a live eel

We exposed 77 adult knifefish to an *Electrophorus varii* (56 cm TL) in an experimental arena. The arena consisted of a 142 L cooler filled with water at 26°C and $13\text{ }\mu\text{S}\cdot\text{cm}^{-1}$ conductivity to match typical stream conditions. Each subject was confined in a plastic mesh tube at the tank center between a pair of recording electrodes. The eel was held loosely in a mesh net on one side of the tank, approximately 30 cm from the knifefish, and acclimated for 15 min before each trial (Figure S4A). The recording methodology for all experiments follows Waddell and Crampton,⁶⁹ with fish signals detected with nickel-chromium tank-end electrodes positioned at the ends

of the recording cooler, amplified using an SR-5113 wide-band amplifier, with 0–30 kHz bandpass filtering, digitized with a USB-6216 analog-to-digital converter and monitored using custom LabVIEW software.

Each knifefish was tested once, with trials conducted in darkness between 18:30 and 00:00 h. No more than five knifefish were tested per night to reduce stress to the eel. During each trial, the eel was lightly prodded with a plastic dowel every ~ 10 s for 30 s to elicit low- and high-voltage EODs, then left undisturbed for 30 s. This one-minute cycle was repeated until a switch-off occurred or the four-minute trial ended. A switch-off was scored when the knifefish ceased EOD production. All knifefish displayed normal behavior following the trials.

Experiment 2

EOD switch-offs of eight knifefish species in response to playback of eel electrolocation EODs

We tested whether the same eight knifefish species used in Experiment 1 exhibited EOD switch-offs in response to playback of low-voltage electrolocation EODs from *Electrophorus varii*. The arena and procedures followed those of Experiment 1, except that the live eel was replaced by a 20 cm playback dipole. Each 30 s playback trial comprised a gradual amplitude ramp followed by 30 s of silence; this cycle was repeated for up to four minutes unless a switch-off occurred earlier. Creation of the playback file is described in “[playback file creation](#),” involving R script 5.⁷²

The positive pole of the playback dipole was positioned 5 cm from the fish’s snout (Figure S4B), simulating an eel approaching head-on. Playback signals were delivered from an Apple MacBook at a 192 kHz sampling rate and 32-bit resolution using the application Decibel v1.3.5, output through an ADI-2 DAC FS digital-to-analog converter (DAC), and passed through a custom-built stimulus isolator that galvanically isolated the playback dipole. The DAC has a built-in amplifier with digital volume control, outputting through a $\frac{1}{4}$ ” TRS headphone port that provides a high-impedance signal with a flat frequency response from 0 Hz to 88 kHz (-1 dB). The headphone output was connected to the stimulus isolator, delivering a floating differential signal across the positive and negative poles. This configuration minimized ground-loop interference and electrical noise while accurately replicating the voltage-source behavior and dipole-like electric field of EODs.^{44,75–77} The isolator used a Jensen JT-6110K-B transformer with a flat frequency response from ~ 0.15 Hz to 120 kHz (-3 dB), reverse-wired to provide a 1:8 voltage step-up to the playback dipole, following Arnegard et al.⁷⁸ The playback electrode comprised a pair of Ag/AgCl wire loops mounted on a PVC rod. The Ag/AgCl surface was produced by chloridizing the silver wire before each experiment via immersion in freshly prepared 5% sodium hypochlorite for 15 min.

At the end of the amplitude ramp, the playback signal’s underwater electric field peak-to-peak amplitude reached a maximum of $19,200 \text{ mV}\cdot\text{cm}^{-1}$, adjusted to match the low-voltage EOD amplitude of the 1.1 m eel used in the range tests. Peak-to-peak amplitude measurements ($\text{mV}\cdot\text{cm}^{-1}$) followed the method of Franchina and Stoddard³⁹ and Waddell and Crampton,⁶⁹ which quantify electric field strength under the load of water as the potential difference across 1 cm, measured 10 cm from the theoretical center of the fish’s dipole-like field in water adjusted to a standardized conductivity ($55 \mu\text{S}\cdot\text{cm}^{-1}$) and temperature (27°C).

The Franchina and Stoddard method requires a calibrated recording setup. A sine-wave calibration signal was output from the headphone output of a computer and routed through a custom-built stimulus isolator incorporating an audio transformer (Radio Shack Audio TX 217-804, with a 1: 6.45 voltage step-up) to a 5 cm playback dipole centered in the recording tank and aligned parallel to the tank sides. A calibration ratio was determined by measuring the playback signal simultaneously across (1) a fixed pair of nickel-chromium tank-end electrodes (‘ref’) and (2) a pair of Ag/AgCl microelectrodes comprising chloridized 1 mm silver beads spaced precisely 1 cm apart, with their midpoint exactly 10 cm from the midpoint of the playback dipole (‘cal’). The playback dipole was then replaced with a live fish (also centered in the recording tank and parallel to the tank sides) and the potential difference of the fish, measured across the tank-end electrodes (in mV), was multiplied by the ref/cal ratio to yield the underwater electric-field amplitude (in $\text{mV}\cdot\text{cm}^{-1}$). Throughout this procedure, signals were amplified using an SR-5113 wide-band differential amplifier, digitized with a USB-6216 analog-to-digital converter, and monitored using custom LabVIEW software.

Experiment 3

EOD switch-off responses of *G. rondoni* to seven playback stimuli

We tested whether *G. rondoni* exhibits EOD switch-offs in response to six types of naturally encountered or manipulated EODs and a playback-of-silence control, summarized in Figures 5A and 5B and Table S6. Design of the playback is described in “[playback file creation](#).” The same arena and playback setup in Experiment 2 were used, except that the positive pole of the playback dipole was positioned 10 cm from the fish’s snout.

We tested 13 *G. rondoni*, each receiving the seven playback types seven times (49 playbacks per individual). Trials were conducted in darkness every other night between 18:30 and 03:00 h over a 28-day period. Following a 10-min acclimation after transfer to the arena, each individual received three or four playback types per night, with 10-min intervals between playbacks. The order of playback types and individuals tested was randomized. For each trial, we recorded the presence or absence of switch-offs, measured playback amplitude at the first switch-off instance during the 30 s ramp, and quantified total switch-off duration.

Playback file creation

For Experiment 2, the goal was to create a noise-free 30-s sequence of low-voltage eel EODs with amplitudes gradually ramping to a realistic maximum. Raw logger recordings of free-swimming eels were unsuitable due to capacitor-induced baseline offsets and uncontrolled body orientation, which alters waveform shape. Instead, we recorded a single low-voltage electrolocation EOD from the

56-cm *E. varii* used in Experiment 1 and assembled a playback sequence by repeating this EOD at a realistic rate obtained from a hunting event, with gradually increasing amplitude. We refer to this playback sequence as the “unmanipulated eel”.

To assemble the 30-s unmanipulated eel playback, we followed seven steps: 1) We recorded a single low-voltage electrolocation EOD using the methods of Crampton et al.⁷⁹ and Waddell and Crampton.⁶⁹ 2) We extracted a natural EOD-rate sequence from a representative 30-s field logger recording. EOD time stamps were extracted using custom LabVIEW software³⁶ employing threshold-crossing and peak detection. 3) In R, we generated a Windows PCM (.wav) file with a 200-kHz sampling rate and 32-bit resolution. At each time stamp (from step 2), we inserted a copy of the EOD from step 1. 4) We rescaled the waveform so that peak amplitude approached the full dynamic range of a 32-bit .wav file. 5) We applied a gradual exponential ramp to simulate eel approach, with the amplitude starting at 3% at 0 s and reaching 100% at 30 s. 6) We down-sampled the sequence in Audacity to 192 kHz for compatibility with the playback hardware. 7) We adjusted the playback through the ADI-2 FS DAC until the maximum underwater electric field strength (at the end of the ramp), measured in water with a conductivity of 55 $\mu\text{S}\cdot\text{cm}^{-1}$ conductivity and temperature of 27 °C, reached 19,200 $\text{mV}\cdot\text{cm}^{-1}$ (at 10 cm). During playback trials, water conductivity in the cooler was lowered to 13 $\mu\text{S}\cdot\text{cm}^{-1}$ to match typical stream conditions.

For Experiment 3, we created five additional 30-s playback sequences with different EOD stimulus types, in addition to the unmanipulated eel sequence and a silence control, resulting in seven total playback types. All sequences, listed below, were assembled using the same general procedure described above. See Table S6 for further parameter details of the stimuli.

1. *Unmanipulated electric eel*. Same playback of low-voltage electrolocation EODs used in Experiment 2.
2. *Manipulated electric eel*. A time-reversed, inverted duplicate of the unmanipulated monophasic eel waveform was appended to each EOD, forming a symmetrical biphasic waveform that reduced spectral power at 30 Hz by 18.2 dB (Figure 5A). To avoid abrupt spectral transitions between the positive and negative phases, we slightly truncated the tail of the positive phase and the leading ramp of the negative phase using Audacity. RMS amplitude was normalized to match the unmanipulated sequence, resulting in a maximum peak-to-peak amplitude of 25,526 $\text{mV}\cdot\text{cm}^{-1}$ at 55 $\mu\text{S}\cdot\text{cm}^{-1}$ and 27 °C.
3. *Natural G. rondoni*. The EODs of a 195-mm adult *G. rondoni* were recorded using the Franchina and Stoddard method.³⁹ A playback was assembled by repeating the EODs at its nighttime rate of ~ 102 Hz. Maximum playback amplitude was adjusted to 130.24 $\text{mV}\cdot\text{cm}^{-1}$ at 55 $\mu\text{S}\cdot\text{cm}^{-1}$ and 27 °C, approximating the measured live fish's 134 $\text{mV}\cdot\text{cm}^{-1}$ value.
4. *Natural B. brevirostris*. The EODs of a 310-mm adult *B. brevirostris* were recorded using the same method. A playback was assembled by repeating the EODs at its nighttime rate of ~ 40 Hz. Maximum playback amplitude was adjusted to 387.51 $\text{mV}\cdot\text{cm}^{-1}$ at 55 $\mu\text{S}\cdot\text{cm}^{-1}$ and 27 °C, approximating the measured live fish's 392 $\text{mV}\cdot\text{cm}^{-1}$ value.
5. *Super-stimulus B. brevirostris*. Amplitude was increased 101-fold (RMS) relative to the natural *B. brevirostris* signal to match RMS amplitude of the unmanipulated eel EOD, resulting in a maximum peak-to-peak amplitude of 32,638 $\text{mV}\cdot\text{cm}^{-1}$ at 55 $\mu\text{S}\cdot\text{cm}^{-1}$ and 27 °C.
6. *Super-stimulus G. rondoni*. We attempted to match RMS amplitude of the unmanipulated eel EOD, but the short *G. rondoni* EOD duration and DAC voltage limits prevented this. Instead, we applied the same 101-fold RMS scaling factor used for *B. brevirostris*, resulting in a maximum peak-to-peak amplitude of 13,465 $\text{mV}\cdot\text{cm}^{-1}$ at 55 $\mu\text{S}\cdot\text{cm}^{-1}$ and 27 °C.
7. *Silence control*. A 30-s .wav file containing only zero amplitude was generated in R and resampled to 192 kHz in Audacity.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical overview

All statistical analyses used generalized linear mixed-effect models (GLMMs) implemented with the glmmTMB package⁸⁰ in R. All tests were two-sided. Model summaries and assumption checks performed with the DHARMA package⁸¹ are provided in the supplemental information and visualized in Data S1. No major violations of model assumptions were detected.

Eel EODs effect on knifefish transit rate

GLMMs 1A and 1B tested the effects of eel EOD presence on knifefish transit rate (Figure 2C). GLMM 1A analyzed *Eigenmannia lor-etana* transit rate (count) in response to eel EOD presence (binary) within the same one-minute recording ($n = 6$ loggers; 106,920 per-minute observations). Logger identity was included as a random effect, and an AR1 autocorrelation term accounted for temporal dependence within each logger. The model used a generalized Poisson distribution with a log link. GLMM 1B analyzed the transit rate of *Gymnorhamphichthys rondoni* using the same structure as GLMM 1A but included an additional covariate, $\log(\text{moonlight} + 0.00007)$, to account for the lunar-phobic response in this species.²² Moonlight refers to ground illuminance (lx) measured with an ILT-5000 low-light-sensitive radiometer (International Light Technologies, Peabody, MA, USA) positioned in a clearing near the stream.²² The small 0.00007 constant permitted log transformation of zero values. Cross-correlation analyses were conducted by creating versions of GLMMs 1A and 1B with time-shifted eel EOD presence predictors (Figure 2C). Predictors were shifted before (−15, −2, −1 min) and after (+1, +2, +15 min) the focal interval while holding other variables constant, thereby testing lead-lag effects of eel EOD presence on knifefish transit rate.

Wild *G. rondoni* switch-offs to live eels

GLMMs 2 and 3 analyzed switch-off responses in wild *G. rondoni*. GLMM 2 examined switch-off occurrence (binary) in response to eel EOD presence (binary) within the same one-minute recording ($n = 8$ individuals; 4,363 observations; Figure 3B). The model used a binomial distribution with a logit link. GLMM 3 analyzed switch-off duration (continuous) in response to eel EOD amplitude (continuous) in the same eight individuals (104 switch-offs, Figure 3C), with both variables $\log_{10}$ -transformed. Eel EOD amplitude refers to the normalized range (0–1) in the logger recording. This model used a Gaussian distribution with an identity link. Individual identity was included as a random effect in both models.

Lab *G. rondoni* switch-offs: Playback responses

GLMMs 4–6 analyzed switch-off responses of laboratory *G. rondoni* to seven playback stimuli (Experiment 3) using a hurdle-model framework. GLMM 4 tested switch-off probability (binary) across seven stimulus types (categorical) (637 observations from $n = 13$ individuals $\times$ 7 trial types $\times$ 7 repeats) (Figure 5C). Using only trials with switch-offs ($n = 7$ individuals; 68 switch-offs), GLMM 5 analyzed playback amplitude at switch-off (continuous), and GLMM 6 analyzed switch-off duration (continuous) in response to manipulated versus unmanipulated eel EOD playbacks. All models included individual identity as a random effect. GLMM 4 used a binomial distribution with a logit link, GLMM 5 a log-normal distribution with a log link, and GLMM 6 a Gaussian distribution with an identity link.